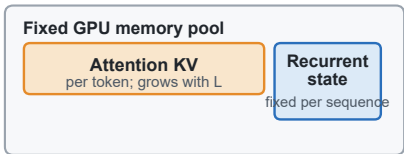


a

Hybrid layer stack



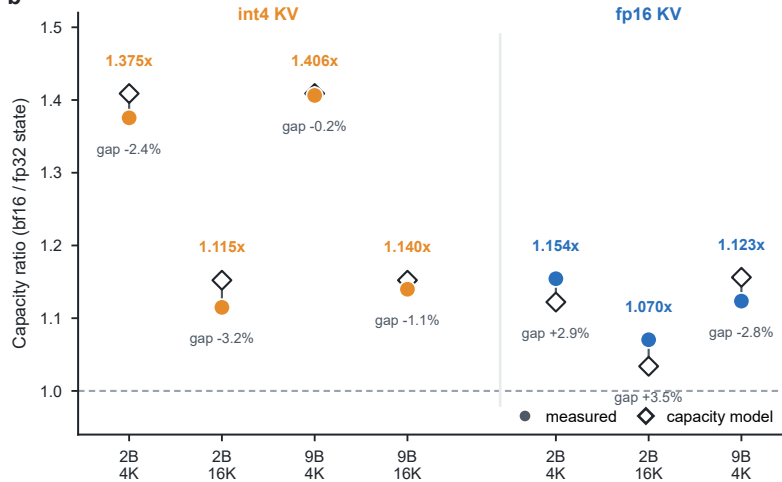
many GDN layers few GQA layers



KV dtype + **state dtype** → **joint budget**

Capacity model $N(L) = M / (A L + G)$; $T(L) = L N(L)$

b



2B - 4K compound KV gain: 2.245x → 2.675x